Sahaj Bhadja

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Summary

Applied AI engineer with 2+ years building **Python**, **LLM orchestration**, **RAG**, and production **API integrations**; experienced translating ambiguous workflows into tested, measurable software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Software Engineering, Database Management Systems, Cloud Computing

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a **multi-tenant AI platform** on Azure SQL and Blob Storage with row-level security for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexed query optimization.
- Restructured fault-tolerant **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs while increasing throughput threefold under concurrent production workloads reliably at scale.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Engineered an AI grading platform using multimodal LLMs and OCR to evaluate 10K+ text and handwritten submissions daily with 90%+ consistency across instructor-facing workflows for 150+ instructors.
- Implemented an AWS-hosted backend serving 150+ instructors, with **REST APIs**, CI/CD-driven rubric workflows, and model-usage tracking that reduced feedback turnaround time by 60% under peak production load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built a GPT-based Decision Transformer and end-to-end **Python evaluation pipelines** processing 100K+ trajectory logs, improving long-horizon behavioral planning by 18% in simulation and closed-loop evaluations at scale.
- Architected a spatial-temporal encoder combining HD-map geometry with ego-agent context through multi-head attention, improving closed-loop planning stability by 15% across simulation evaluations with fewer observed off-road trajectories.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Built cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito while collaborating with Microsoft and GitHub stakeholders across three operating systems.
- Troubleshot **API integrations** and redesigned AWS Glue and S3 ETL workflows, improving response time by 30%, pipeline efficiency by 68%, and maintainability through reusable OOP modules for low-bandwidth users.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Mapped 12+ backend integration points, designed the end-to-end system workflow, and presented architecture to the client, resolving ambiguous requirements before implementation across mobile and web experiences and reducing design errors by 25%.
- Developed Flask APIs and Firebase workflows for 50K+ booking records, delivering sub-200-millisecond responses, real-time status synchronization, and webhook-driven payment automation with 70% less manual handling across cross-platform applications.

Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, C++, Bash
AI & Data: LLM orchestration, RAG, agent workflows, MCP-compatible tools, prompt engineering, deterministic evaluations, PyTorch, OCR, FAISS, Pinecone
Backend & Databases: FastAPI, Flask, REST APIs, microservices, PostgreSQL, SQL Server, Azure SQL, Firebase, Supabase
Cloud & Delivery: AWS, Azure, GCP, Docker, Jenkins, GitHub Actions, CI/CD, Git, Linux
Engineering: API integration, unit testing, integration testing, monitoring, troubleshooting, data modeling, Agile/Scrum, technical documentation

Projects

DealerOpsAgent | Python, MCP-compatible tools, agent graph, deterministic evaluations

- Built and tested a Python agent workflow with MCP-compatible ERP tools, graph-based routing, grounded responses, trace monitoring, feedback capture, and deterministic evaluations for tool selection, safety, and task completion.

Big-Brain | Live | LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP

- Built semantic retrieval and summarization over 50K+ documents using LLaMA-2, FAISS, Pinecone, and REST APIs, maintaining 95% query accuracy while enabling secure collaborative analysis for 200+ users.

Chat2SQL | Code | Java, TypeScript, SQL, Docker

- Built an AI-assisted SQL workspace supporting 10+ database systems, combining schema-aware query generation, validation, and Docker-based deployment to accelerate enterprise data exploration and database management from one unified interface.